ARUN SINGH

arunsin997@gmail.com | +91 743 006 6111 | LinkedIn | GitHub

EDUCATION

Indian Institute of Technology Kharagpur, Bachelor of Technology (Hons.), Computer Science & Engineering 2016 - 2020

EXPERIENCE

Senior Software Development Engineer | *Anakin (YC S21), Bangalore, India* 06/2023 - Present

- Core contributor to anakin.io, an API-as-a-service platform for web scraping. Built backend services, AWS infrastructure, and customer dashboards from zero to production launch.
- Led technical implementation and policy authoring for SOC2 and ISO27001 certification. Scoped controls, implemented encryption, IAM, and audit logging, drove engineering remediation across 20+ services, and owned the auditor engagement.
- Solely architected and operate the anakin.io AWS stack across ECS services, EKS observability, Postgres, Redis, SQS, and a managed Playwright browser pool. Designed for 99.9% availability.
- Architected a config-driven generic ETL framework powering 300 to 400 client-specific pipelines across 15 to 20 B2B customers. Cut new-pipeline setup from a full engineer-day to minutes, saving hundreds of recurring engineering hours every month.
- Built Account Service, a credential pool that lets 30+ scraping services safely share platform accounts with automatic expiry, cooldown, and idempotent release. Underpins authenticated scraping flows that are revenue-critical to the B2B business.
- Designed and operate 20+ stateless scraping services processing 40M+ datapoints/month across the B2B pipelines business, with multi-account rotation, retry, and anti-bot handling.
- Shipped customer-facing and internal dashboards for API key management, credit tracking, traffic monitoring, and scraping controls.

Software Development Engineer | *Amazon Development Centre, India* 11/2021 - 03/2023

- Onboarded the EU region into Amazon's transportation accounting platform end-to-end, owning manifest processing, invoice generation, and book management across the carrier network.
- Led the sole-owner migration of 7 to 8 services from AWS CDK v1 to v2 in line with Amazon's company-wide mandate, coordinating changes across the stack with zero customer-facing downtime.
- Root-caused and fixed a long-standing manifest discrepancy that had been causing book-level accounting mismatches for years, diving into undocumented legacy code that prior owners had tolerated rather than untangled.
- Operated on-call for a high-volume financial system, handling escalations and root-causing production incidents in services with minimal documentation.

Associate Software Development Engineer | *Sigmoid Analytics, Bangalore, India* 10/2020 - 11/2021

- Built data pipelines and dashboard delivery for a global consumer goods company's proprietary recommendation system, automating a data injection and processing workflow that the team believed required dedicated manual engineering. Freed 5 engineers from recurring manual injection work.

SKILLS

Languages: Python, TypeScript, C++

Backend & Data: FastAPI, Flask, Django, Node.js, PostgreSQL, Redis, Pandas, NumPy

Infrastructure & Cloud: AWS (ECS, EKS, EC2, Lambda, S3, RDS, CloudFormation, CloudWatch, EventBridge, SQS), Docker, Kubernetes, Jenkins

Scraping & Automation: Playwright, headless browser infrastructure, anti-bot and session management at scale

Compliance & Security: SOC2 and ISO27001 (led technical implementation and audit)

PUBLICATIONS

- Realistic Multiuser, Multimodal HAR Data Generation through Single User Data Augmentation. ACM/IEEE IPSN 2022.
- Opportunistic Mobility Tracking using Keystroke Patterns for Smartphone Energy Efficiency. IEEE PerCom 2020.